MULTIPLICATION BY D ON AN ABELIAN VARIETY IS FINITE LOCALLY FREE OF DEGREE D^{2 DIM}

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemmas lemma-abelian-variety-projective (0BFA) and more-morphisms-lemma-projective (0B45). cited at proof lines 1993-1995. Existence of the ample invertible module L.
- Lemma properties-lemma-ample-tensor-globally-generated (0890) and lemma-pullbacks-by-n (0BFF). cited at proof lines 1998-2003. Ampleness of N and the identity [d]^*N = N^{(x)d^2}.
- Varieties lemmas 0B5X (or 0BEV) and 02OG. cited at proof lines 2006-2014. Fibres of [d] are zero-dimensional, hence [d] finite.
- Lemma lemma-abelian-variety-smooth (0BFC), algebra lemmas 00R4, 00NQ, varieties 056S, morphisms 02KB. cited at proof lines 2015-2024. Flatness of [d] and finite locally freeness.
- Lemma varieties-lemma-degree-finite-morphism-in-terms-degrees (0BEX). cited at proof lines 2027-2031. The degree identity yielding d^{2 dim(A)}.

Source reference. Original lemma 0BFA: sources/stacks-chow/tags.txt, source lines 14378–14378. Cited at proof lines 1993-1995 to supply the ample invertible module L; load-bearing.

Source reference. Original lemma 0B45: sources/stacks-chow/tags.txt, source lines 13988–13988. Cited at proof lines 1993-1995 alongside 0BFA; load-bearing for the existence of L.

Source reference. Original lemma 0890: sources/stacks-chow/tags.txt, source lines 10480–10480. Cited at proof lines 1998-2000 to see N = L (x) [-1]^*L is ample; load-bearing.

Source reference. Original lemma 0BFF: sources/stacks-chow/tags.txt, source lines 14383–14383. Cited at proof lines 2001-2003 to get [d]^*N = N^{(x)d^2}; load-bearing for the degree formula.

Source reference. Original lemma 0B5X: sources/stacks-chow/tags.txt, source lines 14050–14050. Cited at proof lines 2006-2010 to contradict O_C ample of degree 0; load-bearing for finiteness.

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Source reference. Original lemma 0BEV: sources/stacks-chow/tags.txt, source lines 14363–14363. Cited parenthetically at proof line 2011 as an alternative to 0B5X; navigation for an alternative.

Source reference. Original lemma 02OG: sources/stacks-chow/tags.txt, source lines 3515–3515. Cited at proof lines 2012-2014 to conclude [d] is finite; load-bearing.

Source reference. Original lemma 0BFC: sources/stacks-chow/tags.txt, source lines 14380–14380. Cited at proof lines 2015-2017 to know A is smooth over k; load-bearing for flatness of [d].

Source reference. Original lemma 00R4: sources/stacks-chow/tags.txt, source lines 990–990. Cited at proof lines 2017-2019 to see [d] is flat; load-bearing.

Source reference. Original lemma 056S: sources/stacks-chow/tags.txt, source lines 6720–6720. Cited at proof lines 2019-2021 in the regularity step of the flatness argument; load-bearing.

Source reference. Original lemma 00NQ: sources/stacks-chow/tags.txt, source lines 868–868. Cited at proof lines 2020-2021 in the regularity step; load-bearing.

Source reference. Original lemma 02KB: sources/stacks-chow/tags.txt, source lines 3366–3366. Cited at proof lines 2022-2024 to conclude [d] is finite locally free; load-bearing.

Source reference. Original lemma 0BEX: sources/stacks-chow/tags.txt, source lines 14365–14365. Cited at proof lines 2027-2031 for the degree identity deg(N^{(x)d^2}) = deg([d]) deg(N); load-bearing for the final formula.

Lemma 0.2. *Let k be a field. Let A be an abelian variety over k. Let $[d] : A \to A$ be the multiplication by d. Then $[d]$ is finite locally free of degree $d^{2\dim(A)}$.*

Proof. By Lemma 0BFA (and More on Morphisms, Lemma 0B45) we see that A has an ample invertible module $\mathcal{L}$. Since $[-1] : A \to A$ is an automorphism, we see that $[-1]^*\mathcal{L}$ is an ample invertible $\mathcal{O}_A$-module as well. Thus $\mathcal{N} = \mathcal{L} \otimes [-1]^*\mathcal{L}$ is ample, see Properties, Lemma 0890. Since $\mathcal{N} \cong [-1]^*\mathcal{N}$ we see that $[d]^*\mathcal{N} \cong \mathcal{N}^{\otimes d^2}$ by Lemma 0BFF.

To get a contradiction, let C be a proper curve contained in a fibre of $[d]$. Then $\mathcal{N}^{\otimes d^2}|_C \cong \mathcal{O}_C$ is an ample invertible $\mathcal{O}_C$-module of degree 0 which contradicts Varieties, Lemma 0B5X for example. (You can also use Varieties, Lemma 0BEV.) Thus every fibre of $[d]$ has dimension 0 and hence $[d]$ is finite for example by Cohomology of Schemes, Lemma 02OG. Moreover, since A is smooth over k by Lemma 0BFC we see that $[d] : A \to A$ is flat by Algebra, Lemma 00R4 (we also use that schemes smooth over fields are regular and that regular rings are Cohen-Macaulay, see Varieties, Lemma 056S and Algebra, Lemma 00NQ). Thus $[d]$ is finite flat hence finite locally free by Morphisms, Lemma 02KB.

Finally, we come to the formula for the degree. By Varieties, Lemma 0BEX we see that

$$\deg_{\mathcal{N}^{\otimes d^2}}(A) = \deg([d]) \deg_{\mathcal{N}}(A)$$

Since the degree of A with respect to $\mathcal{N}^{\otimes d^2}$, respectively $\mathcal{N}$ is the coefficient of $n^{\dim(A)}$ in the polynomial

$$n \longmapsto \chi(A, \mathcal{N}^{\otimes nd^2}), \quad \text{respectively} \quad n \longmapsto \chi(A, \mathcal{N}^{\otimes n})$$

we see that $\deg([d]) = d^{2\dim(A)}$. □